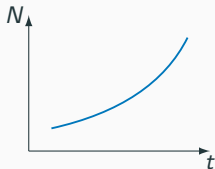


Warm-up: “Winning the War on Poverty”

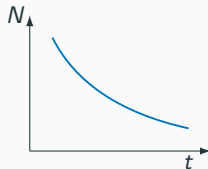
Poll Question

“Winning the war on poverty” has been cynically described as slowing the **rate** at which people are falling below the poverty line. Let N be the number of people in poverty at time t . If we are cynically “winning the war on poverty”, which could be the graph of N ?

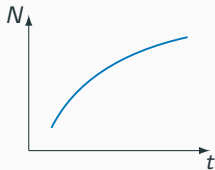
A



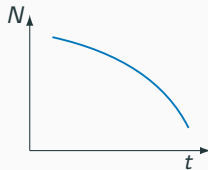
B



C



D



Lecture 4 Average Rates of Change

Big Question:

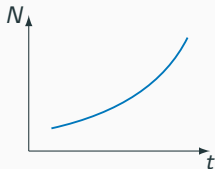
- How can we begin to describe change or **rates of change** for functions that are not necessarily linear?

Warm-up: “Winning the War on Poverty”

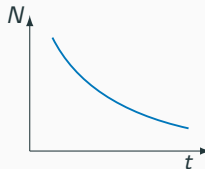
Poll Question

“Winning the war on poverty” has been cynically described as slowing the **rate** at which people are falling below the poverty line. Let N be the number of people in poverty at time t . If we are cynically “winning the war on poverty”, which could be the graph of N ?

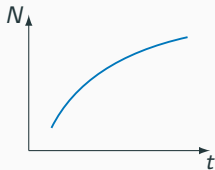
A



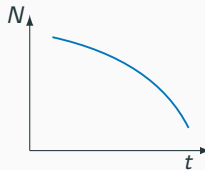
B



C



D



World Population Growth

The world population, N , grew from 3.035 billion in 1960 to 3.701 billion in 1970.

Poll Question

- (a) What was the **total change** in population over this 10-year period?
- (b) Approximately by what **percentage** did the world's population grow over this period?
- A** -22% **B** 22% **C** -0.18% **D** 0.18
- (c) On **average**, by how many people did the population grow **each year** during this 10 year period?

Some Ways of Quantifying Change

Definition

Suppose that $y = f(t)$. As t increases from a to b :

- The (total) **change** in y is

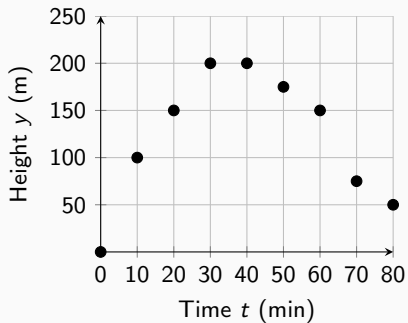
- The **relative change** in y is $\frac{\text{total change in } y}{\text{total change in } y} =$

- The **average rate of change** of y with respect to t is

$$\frac{\text{total change in } y}{\text{total change in } y} =$$

A Hot Air Balloon

An altimeter on a hot air balloon records the following data over a trip:



Interpreting Average Rate of Change

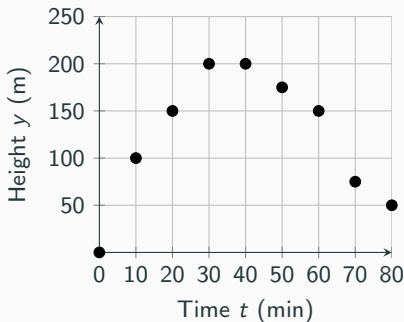
Poll Question

The average rate of change of altitude in the interval $[30, 40]$ is m/min!

Illustrate this on the graph using what we learned last class.

Which of the following is an accurate interpretation of this?

- A** The balloon stays at an altitude of 200 m between 30 and 40 minutes.
- B** The balloon trip is over after 40 minutes.
- C** The balloon rose and fell by equal amounts between 30 and 40 minutes.
- D** The balloon never left the ground.



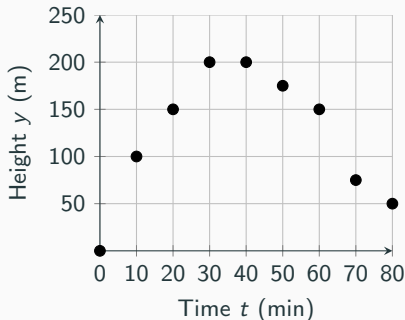
Average Velocity and Acceleration

Poll Question

- (a) Compute the average velocities of the balloon in $[0, 10]$ and $[10, 20]$. Illustrate your reasoning on the graph below.
- (b) Based on this, do you think the hot air balloon's **accelerating** near $t = 10$?

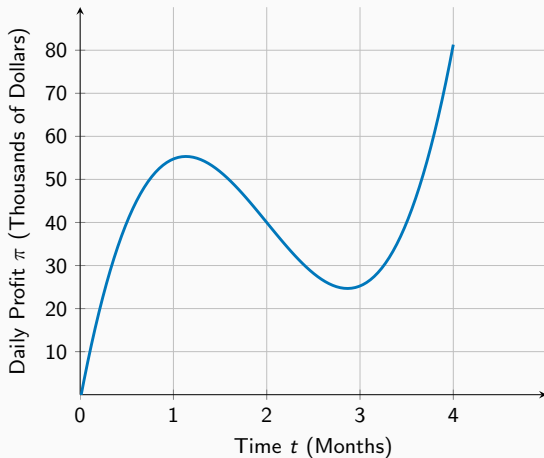
A Yes!

B No!



Daily Profits

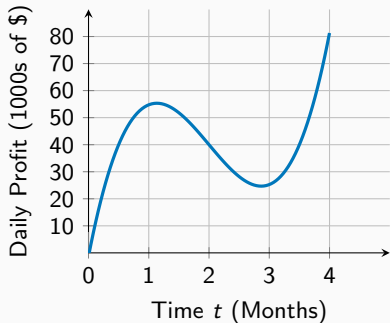
Suppose that the daily profit π for a business as a function of time t in months since the beginning of 2020 is given by the following graph.



Daily profits

Definition

The graph of a function is **concave up** if it bends upward as we move left to right; the graph is **concave down** if it bends downward. A line is neither concave up nor concave down.



Poll Question

The graph of the daily profit is **concave down** over which of the following intervals of time t ?

- ☒ A $[0, 2]$
- ☐ B $[1, 3]$
- ☐ C $[0, 3]$
- ☐ D none of the above

Discussion Question

When the graph is concave down, what do you notice about the average rates of change?

Concavity from a Table

Discussion Question

Consider the function $y = f(x)$ described by the following table:

x	3	4	5	6	7	8	9	12
y	1.2	-1.2	1.9	2.4	3.1	3.8	3.3	2.9

Without sketching a graph, what does the table suggest about the concavity of $f(x)$ around $x = 6$?

To do before next class:

- Please read section 1.3 carefully and do problems 22-30 from HW 1. (HW 1 is currently available on WileyPlus and is due on Tuesday Sep 22 at 10pm.)
- Drop-in hours have started and the Math Learning Centre is open.
- My drop-in hours are 3–4pm on Wednesdays (OISE 5260) and Fridays (OISE 2286).
- Got questions? The Piazza discussion board is now up and running.
- **Reminder:** Tutorials start this week (Thursday/Friday).

Have a great day!

After Class

Label each of the four graphs from the warm-up question as being either:

- Increasing and concave up.
- Increasing and concave down.
- Decreasing and concave up.
- Decreasing and concave down.

Extra Practice!

After Class

Solar panels are arrays of photovoltaic cells that convert solar radiation into electricity. The table shows the annual percent change in the US price per watt of a solar panel.

Year	2007	2008
% growth over previous yr	-3.7	3.6

Did the actual price per watt of a solar panel decrease by 0.1% over the two years listed?

Average Rates of Change: A Graphical Interpretation

What is the relationship between the secant line and the average rate of change?

A demo: <https://www.geogebra.org/m/H3g8nD8f>